

Agentic Cell Erasing System

Vision-guided cell erasing using 3D printer motion

Lab Agents

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The cell erasing task

A request becomes a measured cell selection and an approved erasing path.

“Clean the largest blue cell.”

What the operator needs
A calibrated view, a measured target, and a reviewable erasing path.

What a cell means here
A colored region on the experimental board.

Inspect and measure
Report calibrated coordinates, areas, and distances for blue/green regions.

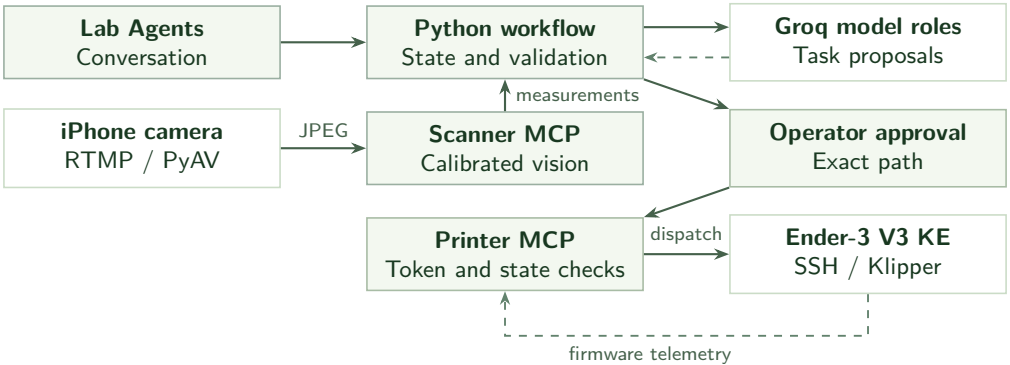
Select within a bounded request
Rank by size, find a nearest region, or check one measured condition.

Prepare and assess
Preview an erasing path, request approval, and compare aligned evidence afterward.

The printer supplies controlled motion to erase selected colored regions on the calibrated board.

System architecture

The workflow combines model proposals, measured evidence, and approval.



MCP uses local stdio tool connections. Cloud calls provide text reasoning and crop-based image assessment.

MCP tools and motion control

Model Context Protocol (MCP) provides a structured tool interface.

Tool interface

`printer_status`	Read machine state
`preview_sequence`	Validate a complete path
`execute_sequence`	Consume the approved token
`emergency_stop`	Request shutdown

Selected printer tools. Requests and results use stdio.

Dispatch checks
Path bounds, homing, start position, offsets, mesh state, heaters, and telemetry as applicable.

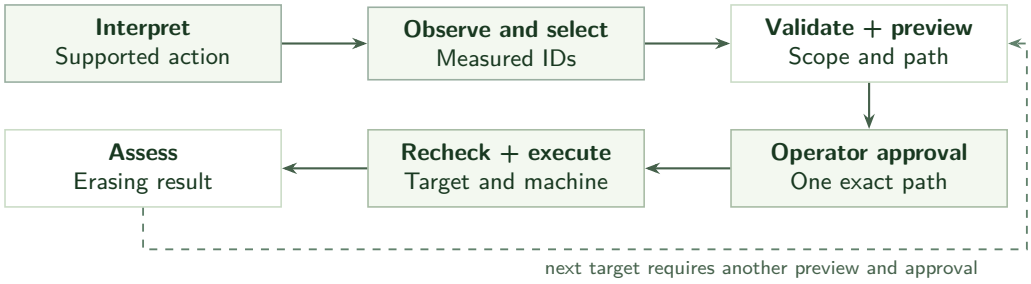
Hardware transport
An authenticated shared SSH session reaches the printer's Klipper Unix socket.

Exposed authority
The lab workflow exposes constrained operations, with no arbitrary G-code, heating, or extrusion.

The configured setup is specific to the Ender-3 V3 KE. Firmware telemetry cannot certify obstacle clearance.

Agent reasoning and the approval loop

Model proposals pass deterministic checks before approved physical action.



Pause conditions

Ambiguous intent, stale or changed evidence, invalid calibration, or expired approval.

Recovery boundary

Uncertain movement or a failed wipe pauses the task. Retrying needs a new task.

Finding cells and checking erasing results

Before erasing, locate the cell. After erasing, check what remains.

Before erasing

Locate the board
Calibration connects the camera view to real positions on the board.

Identify the cell
Vision detects colored cells and measures their positions and sizes.

Check a fresh view
Several recent frames confirm that the target is stable before movement.

After erasing

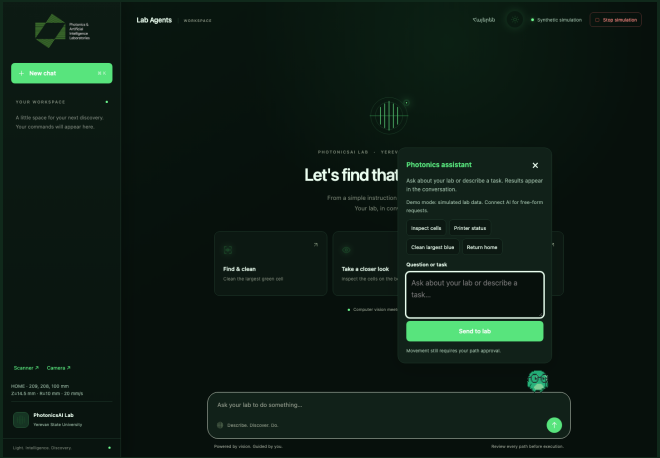
Compare the same area
Align the before and after images to compare the same part of the board.

Check for residue
Assess remaining marks and classify the result as **clean**, **residue**, or **uncertain**.

Accept or pause
Too much remaining color or poor image quality blocks a clean result.

Visual inspection assesses visible marks. Contact pressure and complete erasure need separate validation.

The Lab Agents workspace



Shared conversation
The main composer and mascot panel enter the same workflow.

Lab questions and shortcuts
Status, inspection, and task suggestions prepare a draft for the operator.

Unchanged execution boundary
The operator sends the request and approves each executable path.

UI review image with mocked responses. The local demo uses synthetic observations and a simulated printer.

Example: erasing the largest blue cell

One request links measured selection, approval, execution, and evidence.

01 Observe and select
Acquire fresh calibrated frames and choose the largest eligible blue region by measured area.

02 Review the exact path
Show the target, approach, erasing pass, return route, speed, and limits. Request explicit approval.

03 Revalidate and dispatch
Check the same target and machine start state. Consume the single-use authorization.

04 Assess and record
Compare aligned before/after crops. Record the result or pause on residue or uncertainty.

Configured erasing motion

Approach at clearance height, descend, make one circular pass, lift, and return.

Workflow illustration. No physical outcome is asserted. Initial confirmation expires after 110 seconds.

Evidence and current limits

Software checks, simulation, and hardware observations have distinct limits.

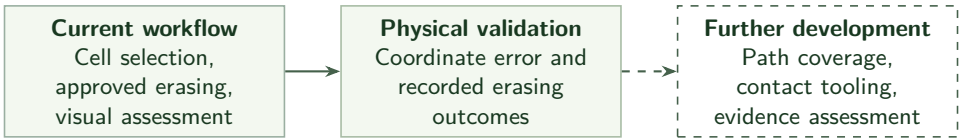
Evidence category	Documented observation	Claim supported
Implemented	Workflow, scanner tools, approval gates, and printer controller in source	Software capability exists
Simulation	94 hardware-free Python tests and 29 browser checks reported, 18 Oct 2026	Software/UI behavior under test
Hardware evidence	Read-only SSH preview and machine-state checks reported, 14 Oct 2026	Specific hardware access and preview
Physical acceptance	New calibrated end-to-end trials remain to be recorded	Outcome reliability pending

Open limitation: the printer handoff records an unresolved live-telemetry fault.

Historical reports were not rerun for this presentation. Physical erasing reliability needs acceptance trials.

Next steps for cell erasing

The next milestone is a measured, supervised validation of the existing loop.



What we will measure

Coordinate error in mm, correct cell selection, and erasing assessment against human labels.

What we will challenge

Stale or changed targets, expired approvals, invalid machine states, and uncertain erasing results.

Discussion: which cell erasing experiment should the lab run first?